



KINEMATION

Retarget Pro

Documentation

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Retarget Pro is an advanced animation tool for Unity, designed to seamlessly transfer your animations from one character to another.

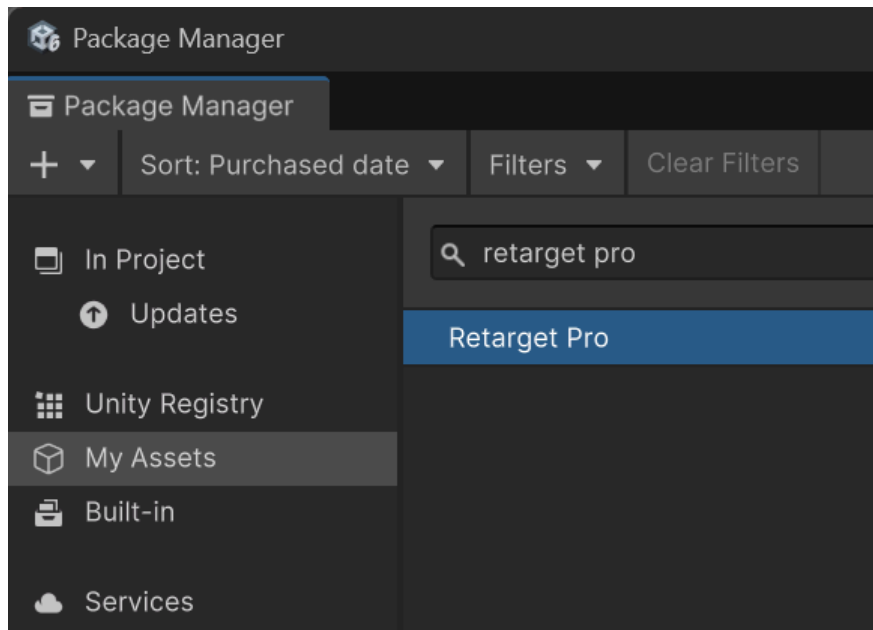
With modular structure and streamlined workflow in mind, this solution provides a higher retargeting quality than [Unity Humanoid system](#), ensuring high-performance for animated characters.

TIP

Note: the tool is able to retarget animations from Humanoid and Generic to another Generic character. Baking animations for Humanoids is **not** supported currently.

Installation

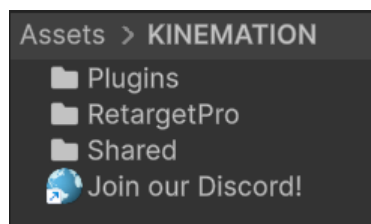
To install **Retarget Pro**, go to Package Manager and search for "Retarget Pro":



TIP

Note: **Retarget Pro** doesn't require any dependencies, and it works with all Unity 6+ versions!

Once package is imported, you'll find Retarget Pro in the "**KINEMATION**" folder:



INFO

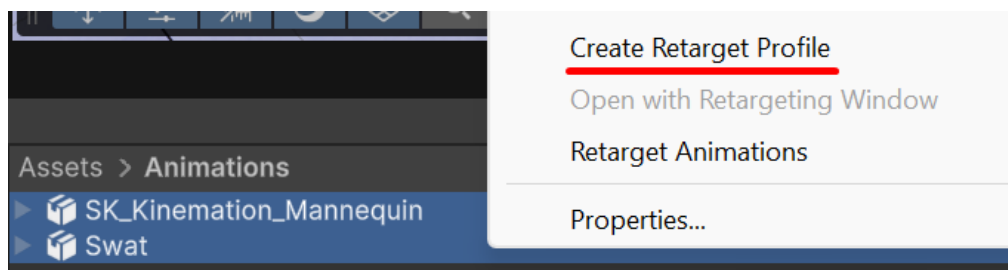
Note: When updating the plugin remove the *KINEMATION* folder first!

Setup

Retarget Pro offers a very easy setup process.

First, select the source character — this is the model to retarget animations FROM. Next, select the target model — this is the skeleton to retarget TO.

Right-click and select "**Create Retarget Profile**":

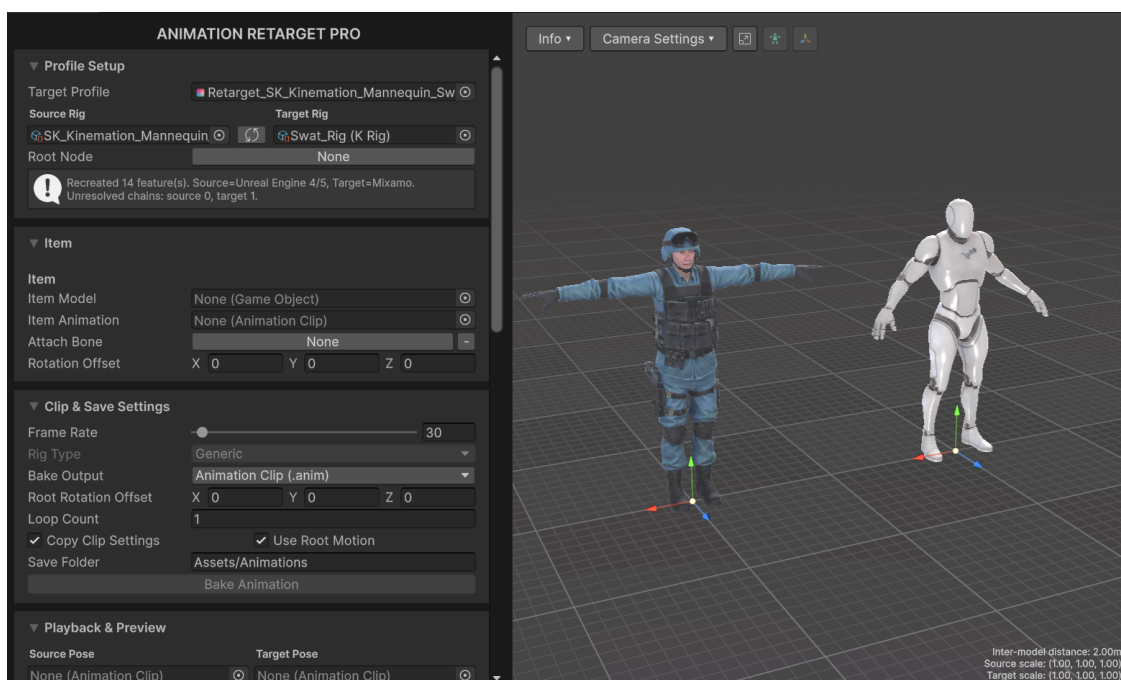


Create a new Retarget Profile.

INFO

Note: *Retarget Profile* is a data asset that contains retargeting settings for your character models. It will be saved to the current folder.

After that, a **Retarget Pro Window** will open up containing previews of selected characters and retargeting settings.



INFO

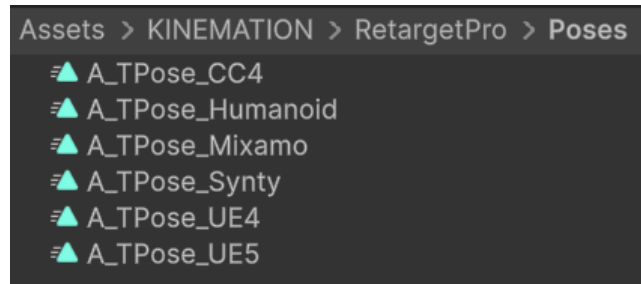
Tip: To find out more about this editor window, see [Retarget Window](#).

Now, we need to assign base poses.

Initial Poses

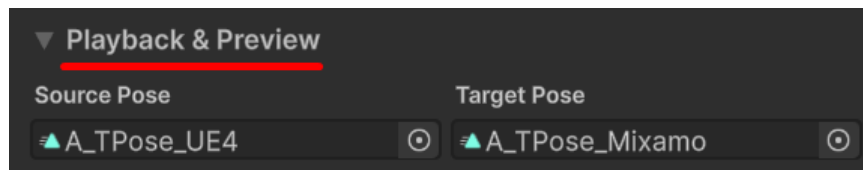
Both characters should have similar poses as this directly impacts retargeting quality.

Retarget Pro comes with T pose Animation Clips for popular skeletons:



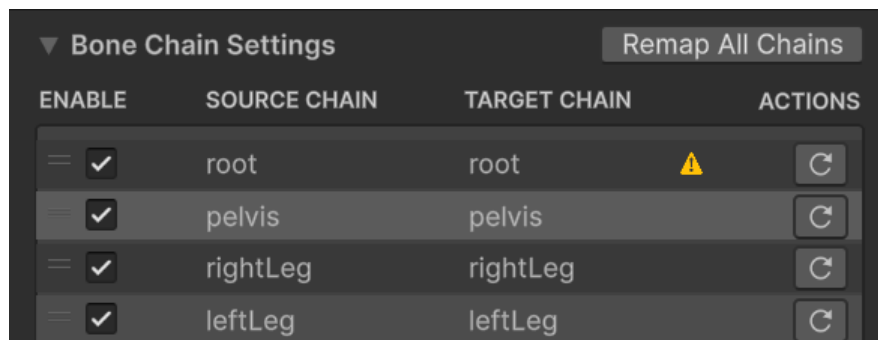
Use these clips as source or target poses.

Assign "Source" and "Target" pose clips if character look different in the preview.



Bone chains

Scroll down to the **Bone Chain Settings** section. This list contains the features that transfer motion from the source chains to the target chains:



Each feature must have a valid **Source Chain** and **Target Chain**. A **None** value or warning icon means that Retarget Pro could not find a matching chain automatically.

INFO

Retarget Pro supports *Mixamo*, *Synty*, *Character Creator*, *UE/UE5*, *Humanoid*, and *Quadruped* skeletons out of the box. Custom bone names, missing rig assets, and different chain structures can still require manual mapping.

What to do if Bone Chains are empty

1 Verify the source and target rigs

Open **Profile Setup** and confirm that **Source Rig** and **Target Rig** reference the correct rigs. The source is the character you retarget animations **from**. The target is the character you retarget animations **to**.

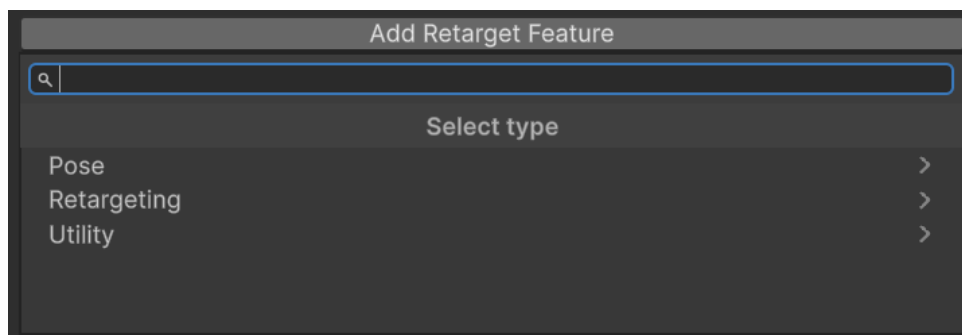
If the rigs are reversed, use the **Swap** button. This swaps the rigs and regenerates the bone chains.

2 Recreate the profile

If the rigs are correct but **Bone Chain Settings** is still empty, click **Remap All Chains**.

3 Click Add Retarget Feature in Bone Chain Settings.

If previous step didn't generate bone chains, click **Add Retarget Feature** in **Bone Chain Settings**.



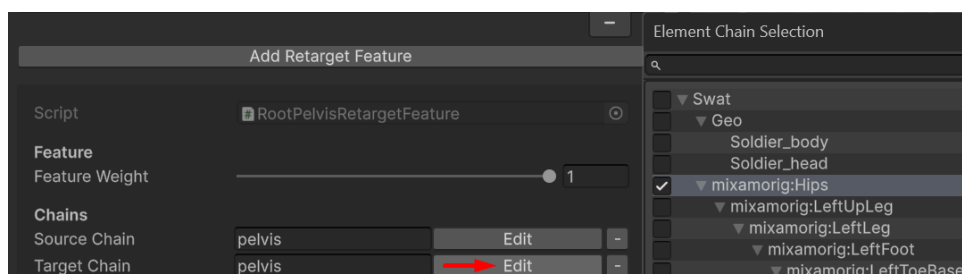
Click Add Retarget Feature to add a feature to the profile.

Select the feature that matches the chain:

- **Root Pelvis Feature** for root and pelvis motion.
- **Basic Feature** for general chains such as the spine, neck, and fingers/toes.
- **IK Feature** for arms and legs.
- **Transform Feature** for a position, rotation, or scale correction.
- **Copy Bone** or **Mirror Feature** for specialized direct-copy or mirrored motion.

4 Map each missing chain

Select a feature with a **None** value or warning icon, then click **Edit**. Select the matching source and target bones in the mapping dialog, then save the mapping. Repeat this for every feature that is incomplete.



Click Edit to assign bones to the selected feature.

Chains

Source Chain

root

Edit

Target Chain

root

Edit

Select the source and target bones for the retarget chain.

TIP

Map the same body region on both sides. For example, map the source `leftArm` chain to the target `leftArm` chain. For **Basic Feature**, keep the source and target chain lengths matched when possible. Arm chains should end at the hand, and leg chains should end at the foot when they are used with **IK Feature**.

Recommended humanoid setup

Use the following as a starting point for a humanoid source or target. Set **Feature Weight** to `1` for primary features. Set it to `0` only when you want to disable a feature without removing it.

Feature	Use-case	Recommended settings
Root Pelvis Feature	<code>root</code> , <code>pelvis</code> bones.	Feature Weight <code>1</code> . Start with Scale Weight and Translation Weight at <code>1</code> when root and pelvis motion should transfer.
Basic Feature	<code>spine</code> , <code>neck</code> , <code>finger</code> chains.	Feature Weight <code>1</code> . Keep Scale Weight at <code>1</code> . Keep Translation Weight at <code>0</code> unless you need to transfer world translation for that chain.
IK Feature	<code>rightArm</code> , <code>leftArm</code> , <code>rightLeg</code> , <code>leftLeg</code> chains.	IK Weight <code>1</code> . Keep Use Chain IK disabled for humanoids. Adjust Pole Weight , Joint Bone , Joint Offset , or Effector Offset only when a limb bends or lands incorrectly.
Transform Feature	Specific correction (e.g., rotation or translation offset).	Add it after the main retargeting features. Use it only for a visible position, rotation, or scale mismatch.

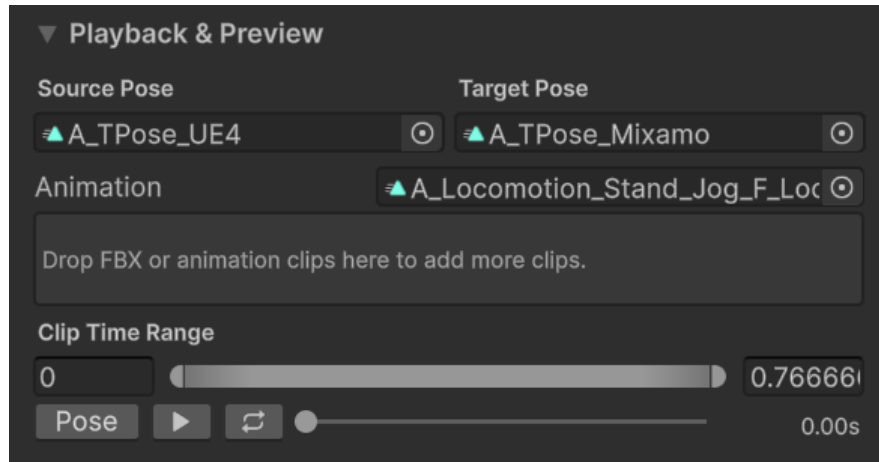
For a typical humanoid, verify the `root`, `pelvis`, `spine`, `neck`, `arm`, `leg`, `toe`, and `finger` chains before previewing an animation. Start with the generated feature order, then make corrections only when the preview shows a problem.

See [Retarget Features](#) for detailed feature settings.

Retargeting

Preview

To start retargeting, go to the **"Playback & Preview"** section and pick an animation:



INFO

Tip: You can drag&drop multiple clips!

Click the "Play", "Loop" button or simply drag the playback slider to retarget animations:



Example retargeting preview.

Adjustments

It's often recommended to adjust retarget features to get the animations look right. Go to the **"Bone Chain Settings"** tab and click on any feature from the list:

▼ Bone Chain Settings				Remap All Chains
ENABLE	SOURCE CHAIN	TARGET CHAIN		ACTIONS
☒	root	root	⚠	↺
☒	pelvis	pelvis		↺
☒	rightLeg	rightLeg		↺
☒	leftLeg	leftLeg		↺

Select a feature.

Once selected, a feature inspector will appear at the bottom of the list:

☒

thumbFingers

thumbFingers

↺

–

Add Retarget Feature

Script

RootPelvisRetargetFeature ⓘ

Feature

Feature Weight

1

Chains

Source Chain

pelvis

Edit

Target Chain

pelvis

Edit

Retarget Settings

Scale Weight

1

Translation Weight

1

Offsets

Offset

X

0

Y

0

Z

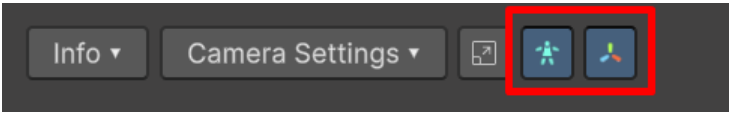
0

Example feature inspector.

INFO

Note: To find our more about each feature settings, please see [Retarget Features](#).

To display the bone renderer and gizmos, toggle the "Skeleton" and "Gizmo" icons in the scene preview:



The bone renderer will display source and target bone chains, and gizmos will expose transform handles to adjust bone positions.



FPS Animations

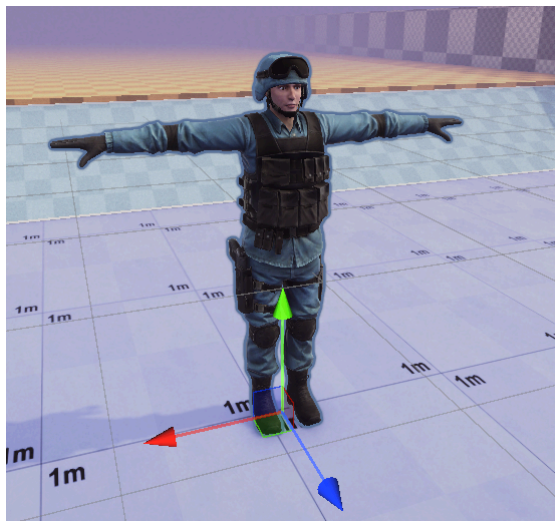
Retarget Pro has a special [FPS Retargeting](#) feature that can be used to share first-person weapon animations.

Weapon Bone

FPS Retargeting relies on a weapon bone — a special game object in the hierarchy that will be used to bake weapon movement.

If your skeleton doesn't have a gun bone, you can add it manually:

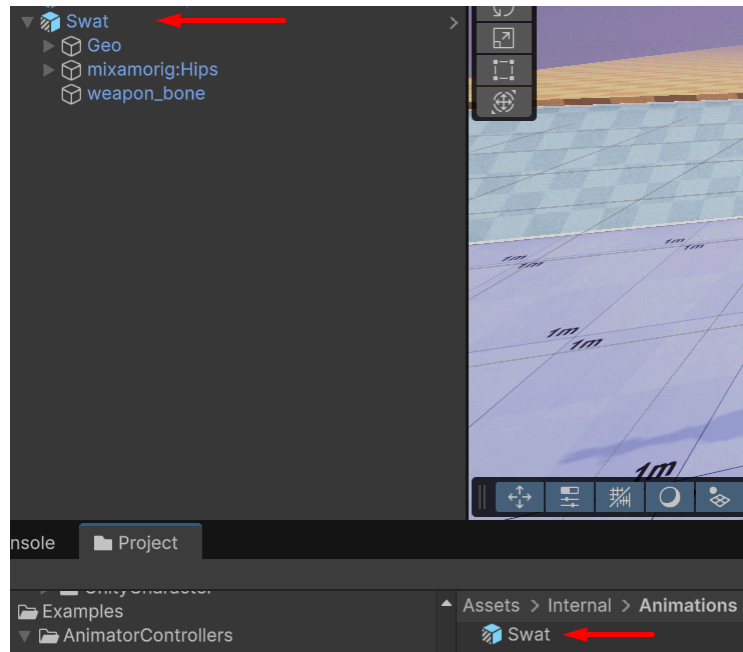
1. Drag your model into the scene



2. Add an empty "weapon_bone" Game Object.

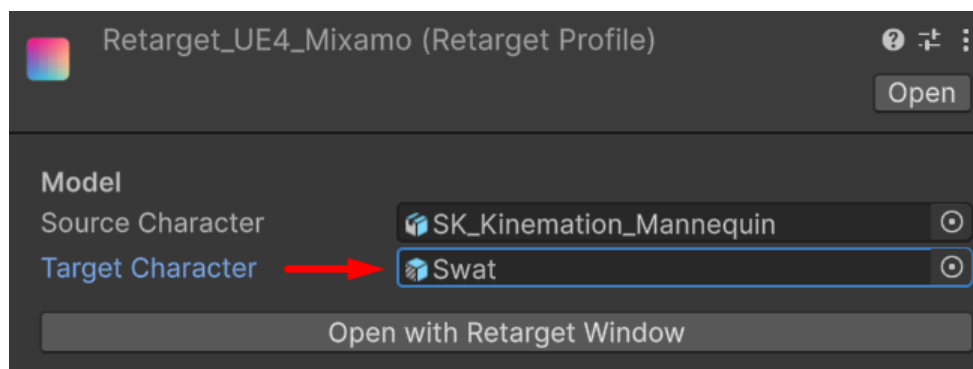


3. Save your model as a prefab variant.



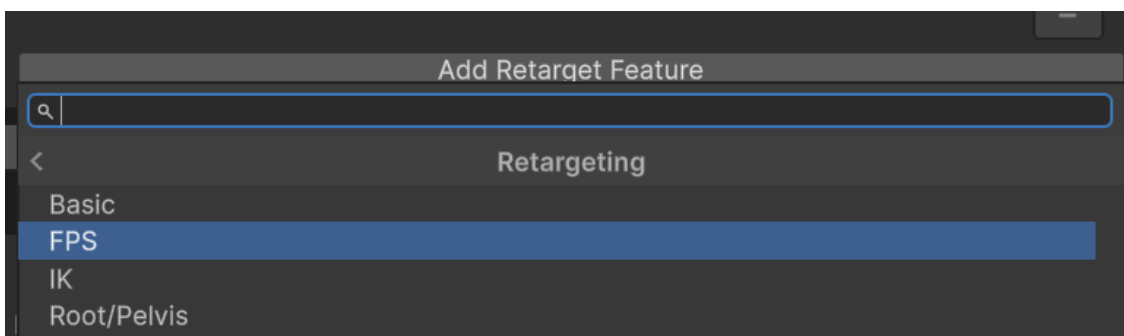
4. Update the character model.

Go to the Retarget Profile asset (or create a new one) and assign the new prefab variant:



Setup

Open the [Retarget Window](#) and go to the "Bone Chain Settings" and add an FPS feature:



Select "FPS".

Place this feature after the arms and before finger retargeting features:

☒	neck	neck
☒	rightArm	rightArm
☒	leftArm	leftArm
☒	FPS Animation Retarget	
☒	indexFingers	indexFingers

For most characters, bone chains will be mapped automatically. It's recommended to double-check by manually editing source and target chains:

Source Chains	
Source Right Arm	Edit Source Right Arm
Source Left Arm	Edit Source Left Arm
Source Weapon	Edit Source Weapon
Target Chains	
Target Right Arm	Edit Target Right Arm
Target Left Arm	Edit Target Left Arm
Target Weapon	Edit Target Weapon

Weapon Preview

It's possible to add a weapon preview directly to the character.

Go to the **"Item"** tab and assign the weapon model, animation and the attach bone:

▼ Item

Item

Item Model

SKM_AK105

Item Animation

A_W_AK105_Reload_Tac

Attach Bone

weapon_bone

Rotation Offset

X 0 Y 0 Z 0

INFO

Tip: This will help with hands positioning.

Retargeting

Finally, select the animation you'd like to retarget and click "Play":



Select "FPS Animation Retarget" to activate gizmos.

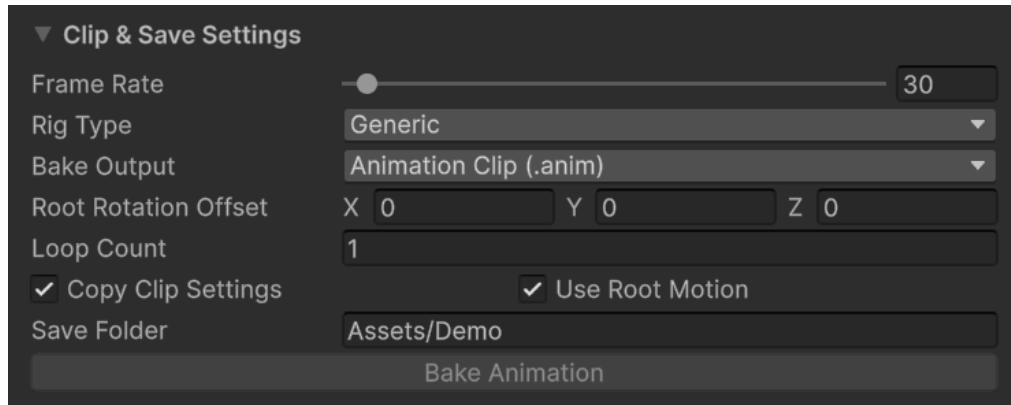
INFO

Tip: Click IK target cubes to activate Gizmo handles to position the weapon, hands and elbows.

Bake & Export

Standard Export

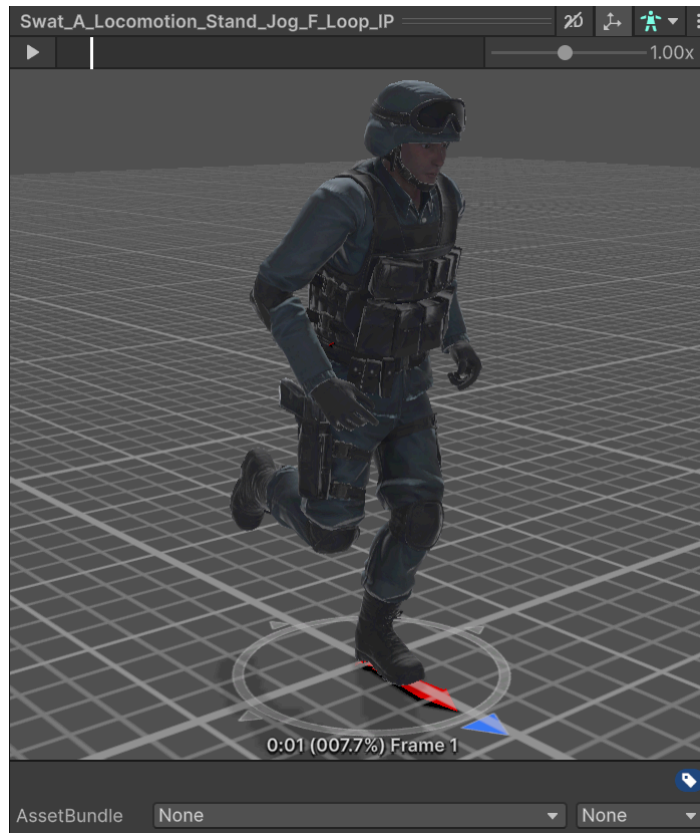
It's possible to bake retargeted animations directly in the **Retarget Pro Window**.



Clip & Save Settings.

- **Frame Rate** — defines the frame rate, in frames per second (FPS), for exported animations.
- **Rig Type** — defines the target character rig type: **Generic** or **Humanoid**.
- **Bake Output** — specifies the export format. **Animation Clip** and **FBX** are supported. **FBX** is recommended.
- **Root Rotation Offset** — applies a rotation offset to the root bone in baked animations.
- **Loop Count** — defines the number of animation loops to include when baking. This is useful for looping locomotion animations, such as walking or running.
- **Copy Clip Settings** — determines whether clip settings, such as **Loop** and **Bake Into Pose**, are copied from the source animations.
- **Use Root Motion** — controls whether root motion tracks are included in the baked animations.
- **Save Folder** — specifies the folder where exported animations are saved.

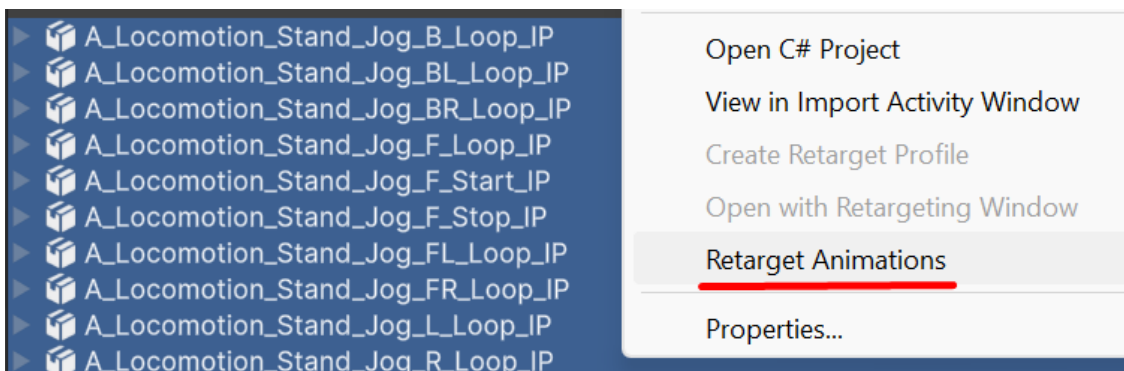
To bake animations, click **Bake Animation**. Once baking is complete, the animation is saved to the selected folder.



Successfully exported animation.

Batch Export

To quickly retarget and bake multiple animations, select the animations, right-click the selection, and choose **Retarget Animations**.



Assign a **Retarget Profile**, configure the export settings, and click **Bake Animations**.

Retarget Animations

▼ Profile Setup

Target ProfileRetarget_Mocap (Retarget Profile)

▼ Clip & Save Settings

Frame Rate30

Rig TypeGeneric

Bake OutputFBX (.fbx)

!

Uses KINEMATION ASCII FBX exporter package.

Root Rotation Offset

X0Y0Z0

Loop Count1

☒ Copy Clip Settings☒ Use Root Motion

Save FolderAssets

▼ Batch Retargeting

!

7 clips selected.

Retarget Animations

Dynamic Retargeter

The **Dynamic Retargeter** transfers the current pose of one character to another at runtime. Use it when you want a target character to follow a source character without baking a new animation clip.

INFO

Before you start, install **Retarget Pro** and create a [Retarget Profile](#) that maps the source rig to the target rig.

What you need

- A source character that produces an animation pose at runtime.
- A target character that will receive the pose.
- A **Retarget Profile** for the two rigs.
- An **Animator** component on the target character. Leave its **Controller** field empty.

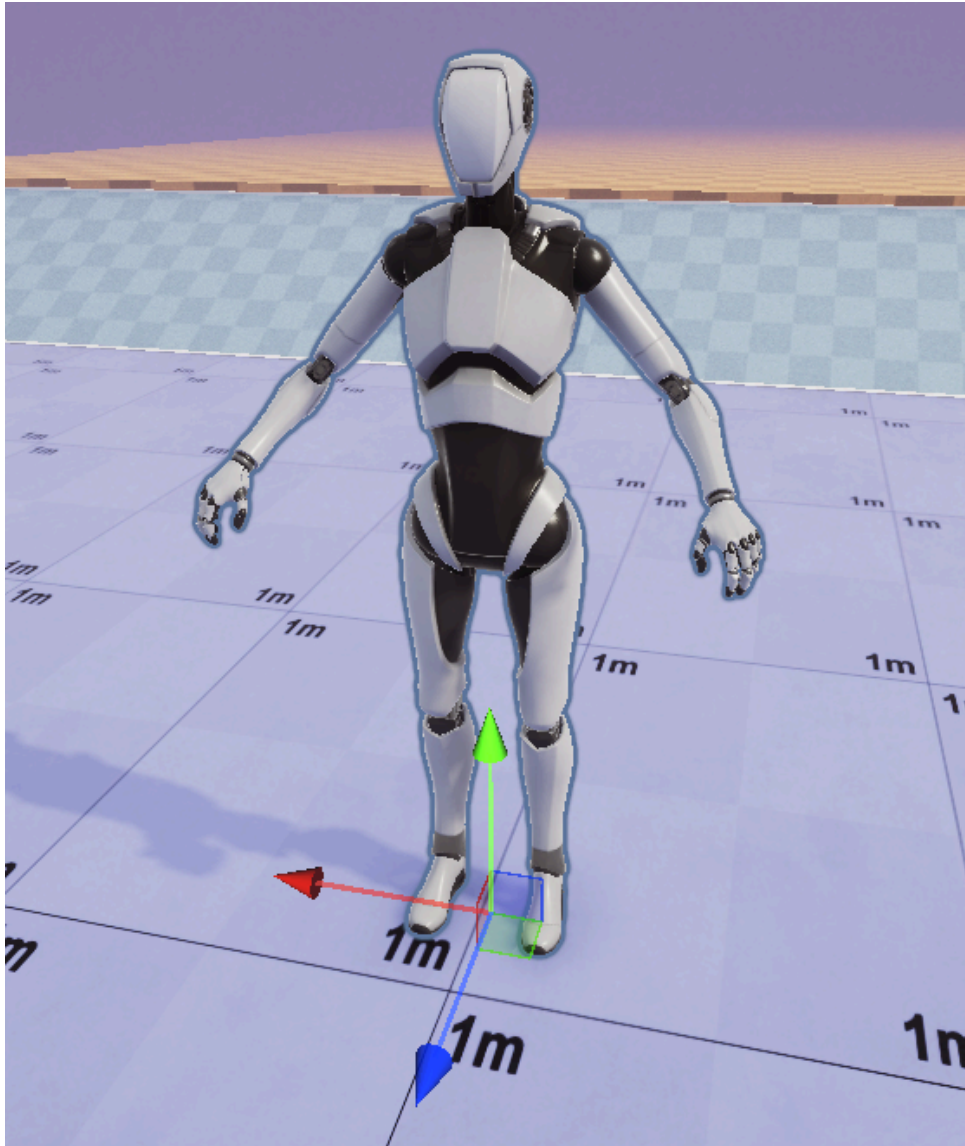
TIP

The source character must stay active so its animation can be evaluated. You can hide its mesh after setup, but do not disable the source GameObject.

Set up dynamic retargeting

1 Add the source character

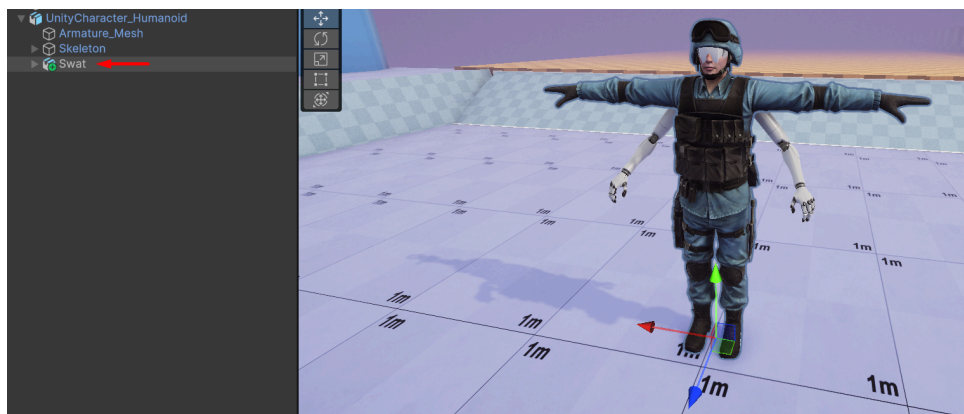
Add the character that contains the animation you want to use to the scene. Confirm that its **Animator** or other pose provider is enabled and produces the expected pose in Play mode.



Add the source character to the scene.

2 Add the target character under the source

Add the target character to the source character's hierarchy. This keeps both characters together in the scene while the source pose is transferred.

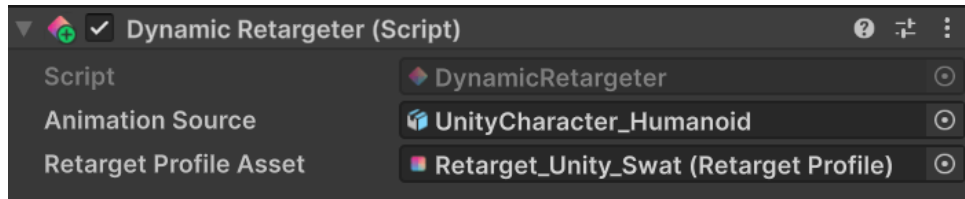


Add the target character to the source hierarchy.

3 Add and configure Dynamic Retargeter

Select the target character and add the **Dynamic Retargeter** component. Assign the source character to **Source Game Object**, then assign the **Retarget Profile** that maps the two rigs.

Make sure the target character has an **Animator** component. Do not assign an **Animator Controller** to this component; the Dynamic Retargeter supplies the target pose.



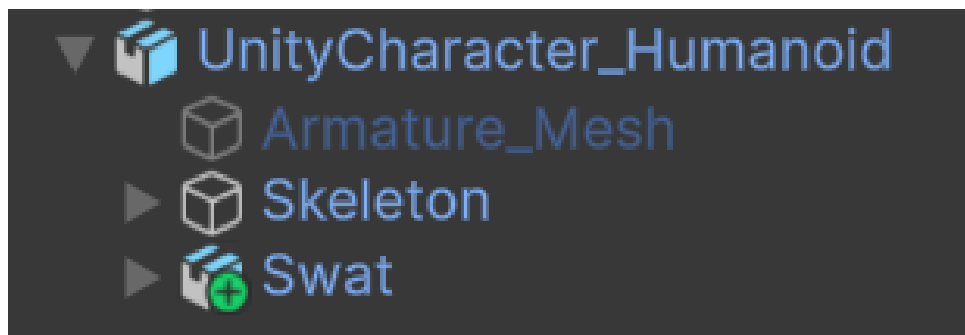
Add Dynamic Retargeter to the target character and assign its references.

4 Hide the source mesh

Hide the source character's mesh so you only see the retargeted target. Disable the source renderers or mesh objects, but keep the source GameObject and its animation components enabled.

WARNING

Do not deactivate the source GameObject. The target needs the source pose to be evaluated every frame.



Hide the source mesh while keeping the source character active.

5 Enter Play mode

Enter Play mode and play an animation on the source character. The target character should follow the source pose in real time.

CHECK

Dynamic retargeting is working when the target follows the source animation while the source mesh remains hidden.



The target character follows the source animation at runtime.

Improve the result

If the target follows the animation but the pose is incorrect, check the profile before adding offsets in your own scripts.

- Assign compatible **Source Pose** and **Target Pose** clips in the Retarget Profile. Similar initial poses produce more reliable results.
- Open the profile and check the **Bone Chain Settings** list for missing or warning-marked chains. Edit the affected source and target chains manually when automatic mapping is incomplete.
- Confirm that the source character is the first rig in the profile and the target character is the second rig. A reversed profile maps the animation in the wrong direction.
- Make sure the target hierarchy has not changed after the profile was created. Recreate or update the profile when the rig structure changes.

INFO

Use the [Retarget Window](#) to inspect profile mappings, pose clips, feature order, and the **Notifications & Help** tabs before testing the runtime component again.

Troubleshooting

The target does not move

Check these references in order:

1. The target **GameObject** is active and has an enabled **Animator**.
2. The target **Animator** has no **Controller** assigned.
3. **Source Game Object** points to the animated source character.
4. **Retarget Profile** points to a valid profile for the two rigs.
5. The source **GameObject** and its **Animator** are active and producing a pose.
6. The Dynamic Retargeter component is enabled.

WARNING

If the source character is disabled, or if you hide it by disabling its whole **GameObject**, the Dynamic Retargeter has no source pose to copy.

Only some bones follow

The profile is missing mappings for the chains that do not move. Open **Bone Chain Settings**, select the affected feature, and assign the correct source and target chains. See [Retarget Features](#) for feature-specific settings.

The target is offset or rotated incorrectly

Check the source and target pose clips first. Then check the root and pelvis mappings, bone-chain order, and the local orientation of the source and target rigs. Fix the profile mapping before compensating with runtime transform offsets.

The source character is visible behind the target

Hide the source mesh renderers or mesh objects, not the source **GameObject**. The source hierarchy and animation components must remain active for the target to update.

The target jitters or is overwritten

Another **Animator**, **IK system**, **physics component**, or **script** may be writing to the target after the retargeting pass. Temporarily disable other systems to find the conflict, then make sure the source pose is evaluated before the final procedural corrections.

Changes do not appear after changing the profile

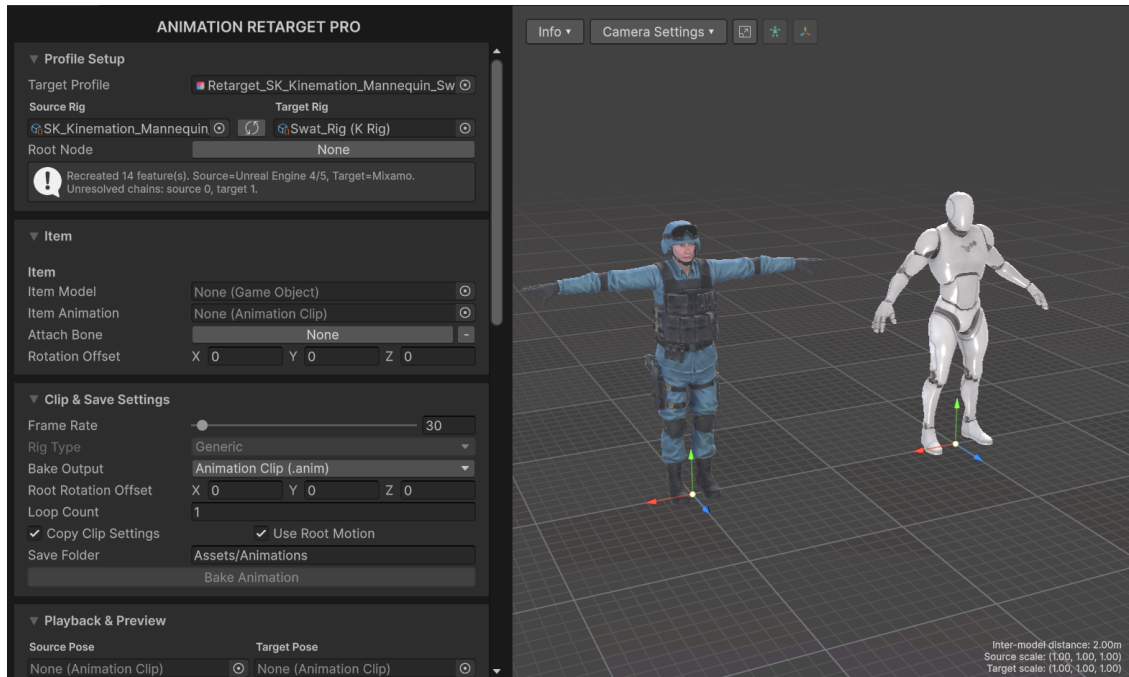
Stop Play mode, verify the **Source Game Object** and **Retarget Profile** references, and start Play mode again. If the rig hierarchy changed, update the profile mappings before testing.

TIP

Dynamic Retargeter evaluates a source pose at runtime. It does not create a new Animation Clip. Use [Bake & Export](#) when you need a reusable baked animation asset instead.

Retarget Window

Retarget Pro Window is the main workspace for adjusting, previewing and baking animations.



It's opened automatically when creating a new Retarget Profile.

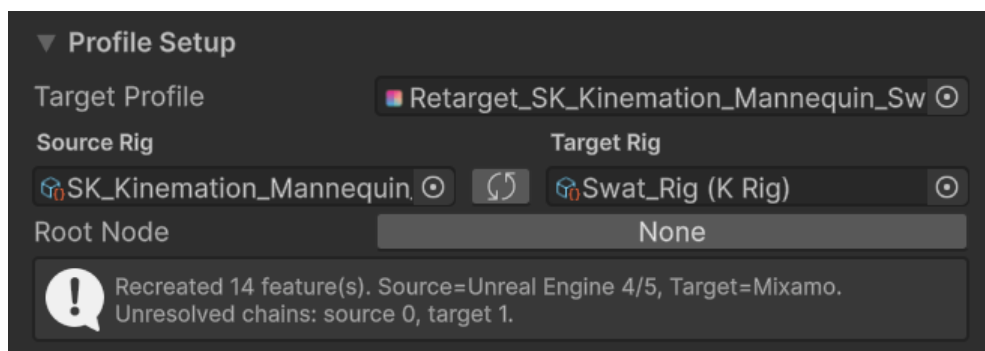
To open it manually, double-click on a Retarget Profile or go to **Window** ► **KINEMATION** ► **Retarget Pro**.

Control Panel

This is the left side of the window that contains various retargeting settings, grouped as dropdown tabs.

Profile Setup

This tab displays general character skeleton settings.



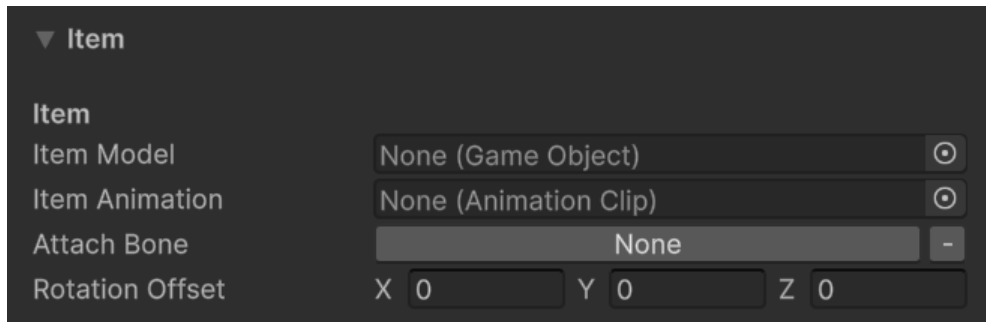
- **Target Profile** — current Retarget Profile.
- **Source Rig** — skeleton of the source character.
- **Target Rig** — skeleton of the target character.
- **Root Node** — the top/root bone in the hierarchy (only used for root motion).

Press the "swap" button to swap source and target character skeletons. This action will re-generate bone chains for both skeletons.

Root Node is to properly bake root motion. If your target character has a root bone (e.g., root, skeleton, armature, etc.) specify it here.

Item

This tab displays settings for a preview item (e.g., weapon).



- **Item Model** — the skinned mesh of the preview item (e.g., rifle or a sword).
- **Item Animation** — the animation to preview on the selected item (e.g., reload).
- **Attach Bone** — the bone to attach the item to.
- **Rotation Offset** — local rotation offset applied to the item.

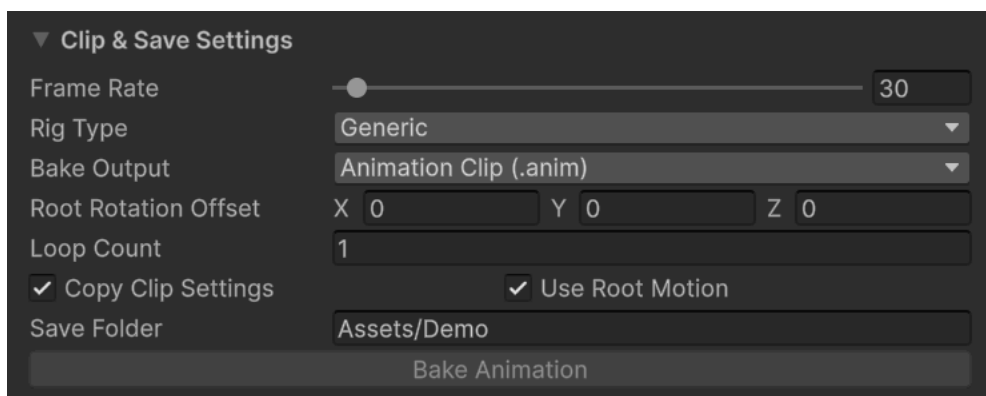
These settings are useful when you want to preview an item together with the character animation.

INFO

Example: When retargeting a reload animation, it's useful to have a weapon animated to see how to position character IK.

Clip & Save Settings

This tab displays animation exporting settings.

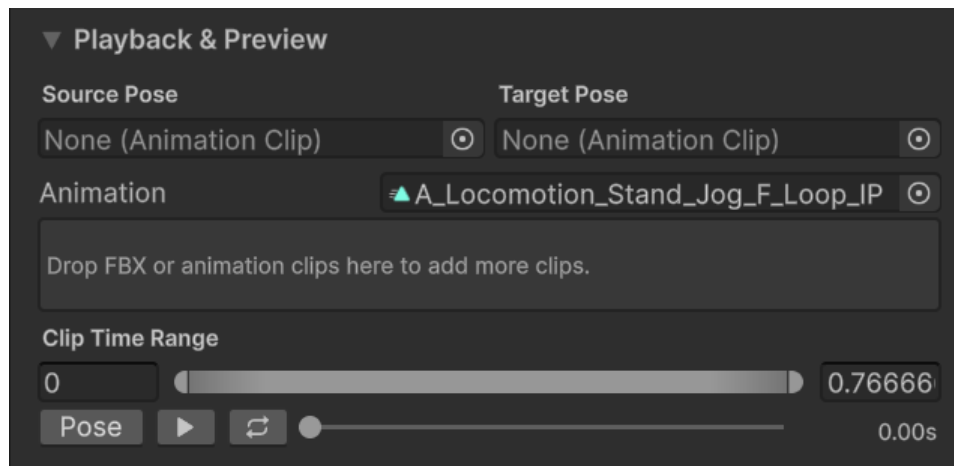


- **Frame Rate** — target export frame rate.
- **Rig Type** — both Generic and Humanoid rigs are supported.
- **Bake Output** — bake the animation as a clip or FBX file.

- **Root Rotation Offset** — local rotation offset for the root bone. Useful for models with non-standard/offset roots.
- **Loop Count** — number of animation loops to include. Useful for continuous locomotion actions.
- **Copy Clip Settings** — whether to copy source clip settings (e.g., Loop, Bake Into Pose, etc.).
- **Use Root Motion** — whether to use root motion from the source character.
- **Save Folder** — output directory for retargeted animations.

Playback & Preview

This tab displays the playback and what animations to use.



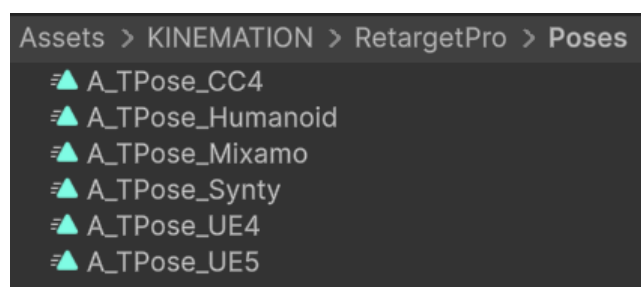
- **Source Pose** — source character base pose (e.g., T or A pose).
- **Target Pose** — target character base pose (e.g., T or A pose).
- **Animation** — a single or multiple clips to retarget.
- **Clip Time Range** (for single clips) — defines the start and end animation time.
- **Pose** — stops the playback and resets character poses.
- **Play** — starts the retargeting.
- **Loop** — toggles animation looping.

Source and **Target Pose** clips are sampled before the retargeting starts.

INFO

Tip: Initial poses directly impact the retargeting quality, so it's important that both characters have identical poses.

Retarget Pro comes with T poses for popular skeletons:



Use these clips as initial poses.

Bone Chain Settings

This tab displays what retarget features and chains are used in this profile.

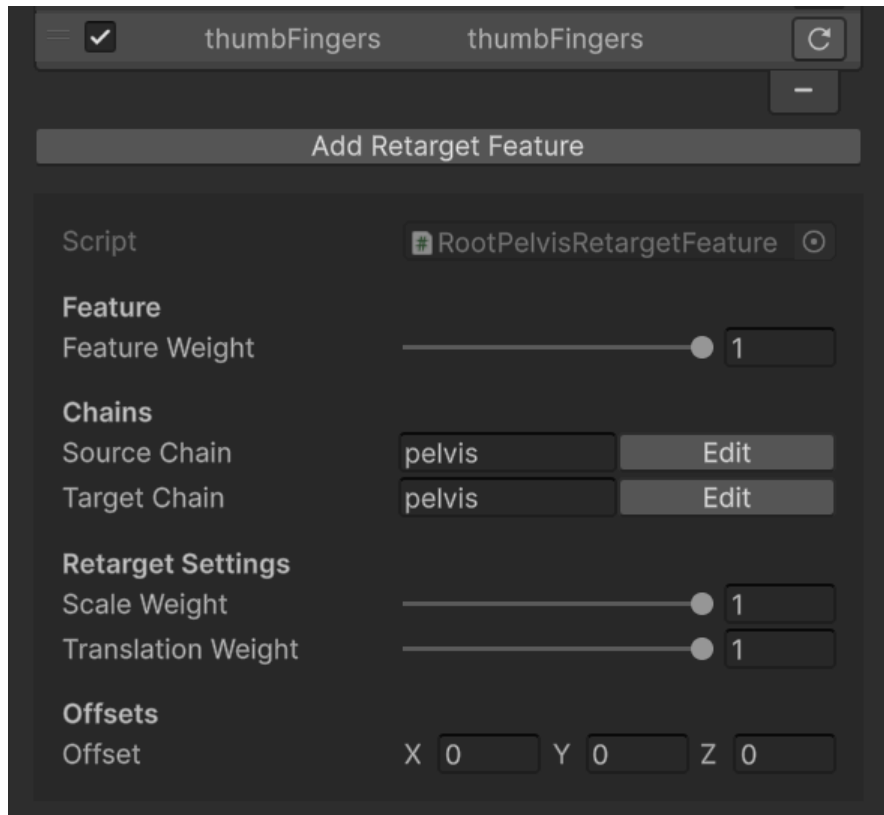
▼ Bone Chain Settings				Remap All Chains
ENABLE	SOURCE CHAIN	TARGET CHAIN	ACTIONS	
☒	root	root		
☒	pelvis	pelvis		
☒	rightLeg	rightLeg		
☒	leftLeg	leftLeg		
☒	toes	toes		
☒	spine	spine		
☒	neck	neck		
☒	rightArm	rightArm		
☒	leftArm	leftArm		
☒	indexFingers	indexFingers		
☒	middleFingers	middleFingers		
☒	ringFingers	ringFingers		
☒	pinkyFingers	pinkyFingers		
☒	thumbFingers	thumbFingers		
Add Retarget Feature				

All features and bone chains are displayed as a single list. The order matters — features are applied as they are ordered in the list.

INFO

Tip: See [Retarget Features](#) to find out more about all retarget features.

To see feature settings, click on any list elements and scroll down:



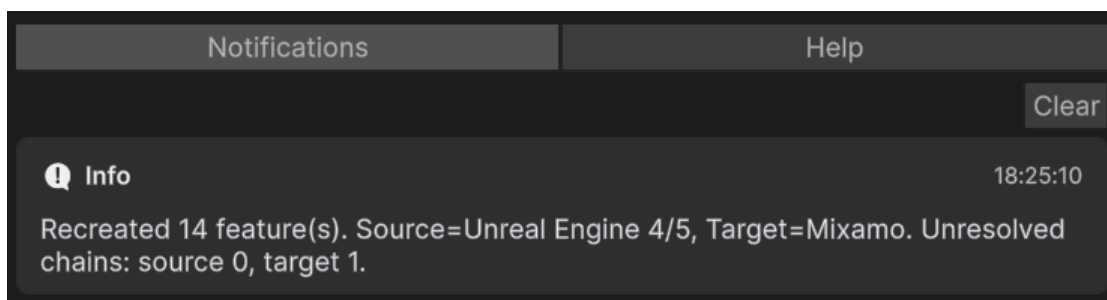
Retarget Feature inspector.

Retarget Pro natively supports popular skeleton types:

- Mixamo.
- Synty.
- Character Creator.
- UE4/UE5.
- Humanoid (General)
- Quadruped (Animals).

Notifications & Help

These last tabs are used to display notification and help messages, which can be quite helpful with troubleshooting.



Retarget Features

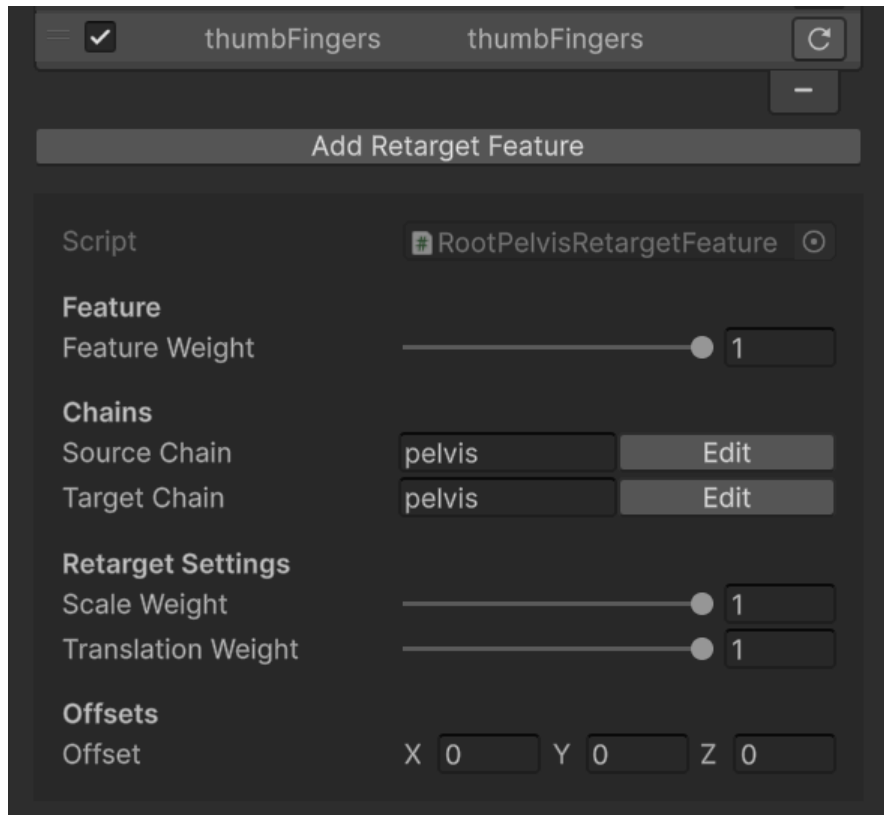
Retarget Feature is a code module that retargets motion from one bone chain to another.

These features are displayed as a list in the Retarget Window:

=	☒	root	root	
=	☒	pelvis	pelvis	
=	☒	rightLeg	rightLeg	
=	☒	leftLeg	leftLeg	
=	☒	toes	toes	
=	☒	spine	spine	
=	☒	neck	neck	
=	☒	rightArm	rightArm	
=	☒	leftArm	leftArm	
=	☒	indexFingers	indexFingers	
=	☒	middleFingers	middleFingers	
=	☒	ringFingers	ringFingers	
=	☒	pinkyFingers	pinkyFingers	
=	☒	thumbFingers	thumbFingers	

Humanoid retarget features.

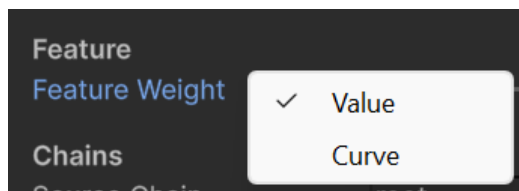
To view feature settings, click on it and scroll down:



Retarget Feature inspector.

Feature Weight controls how strongly a feature is applied. A value of **0** disables the feature, while **1** applies it at full strength.

Right-click **Feature Weight** to select the alpha mode:



- **Value** — the default float slider. Recommended for retargeting features that should remain active throughout the entire animation.

Curve — controls the feature weight using an **Animation Curve**. Useful for features that should only be active during specific parts of an animation.

■ Basic Feature

■ Copy Bone

■ IK Feature

■ Root Pelvis Feature

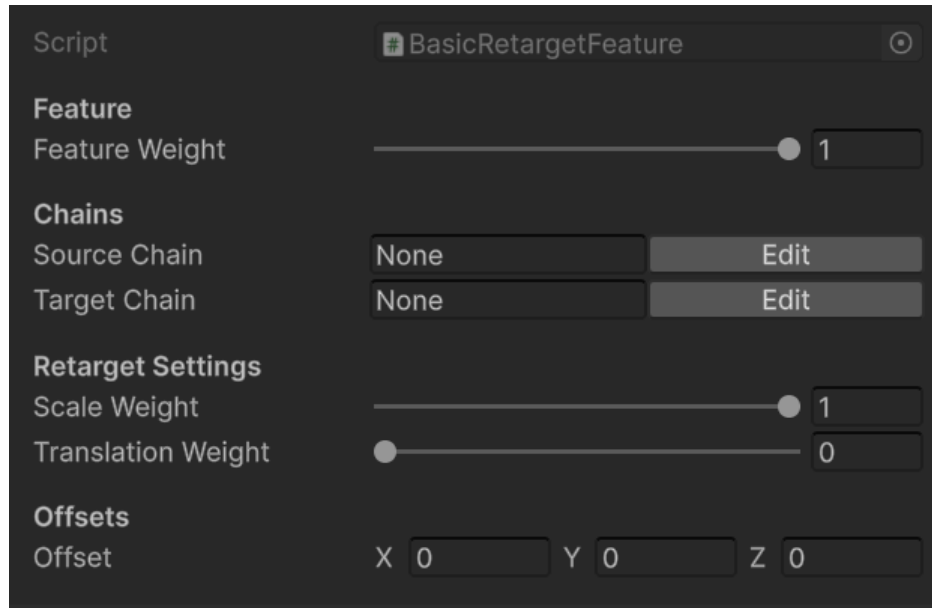
■ **Mirror Feature**

■ **Transform Feature**

■ **FPS Retargeting**

■ Basic Feature

Basic Retarget Feature is used for simple bone chains, like spine or neck.

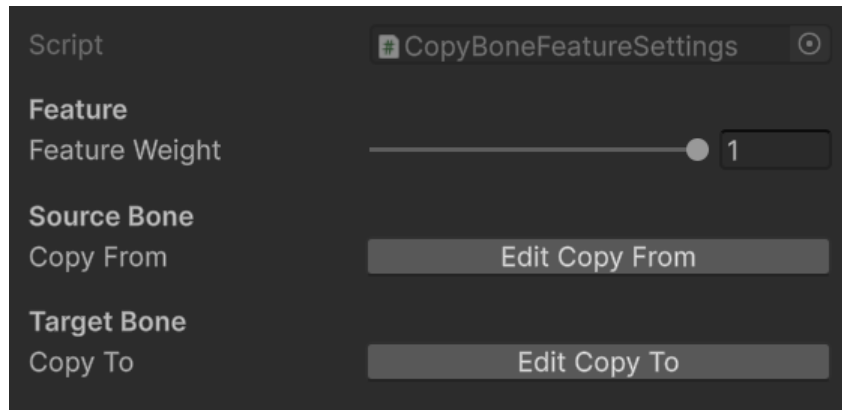


Basic Retarget Inspector.

- **Feature Weight** — Controls the influence of this feature.
- **Source Chain** — Source bones to retarget from.
- **Target Chain** — Target bones to retarget to.
- **Scale Weight** — Controls the influence of the body scale factor. If set to 1, it will be fully applied.
- **Translation Weight** — Controls the influence of the translation factor.
- **Offset** — Translation offset property.

■ Copy Bone

Copy Bone Feature directly copies motion from one bone to another. It can be useful when a motion has to be copied directly from one skeleton to another.

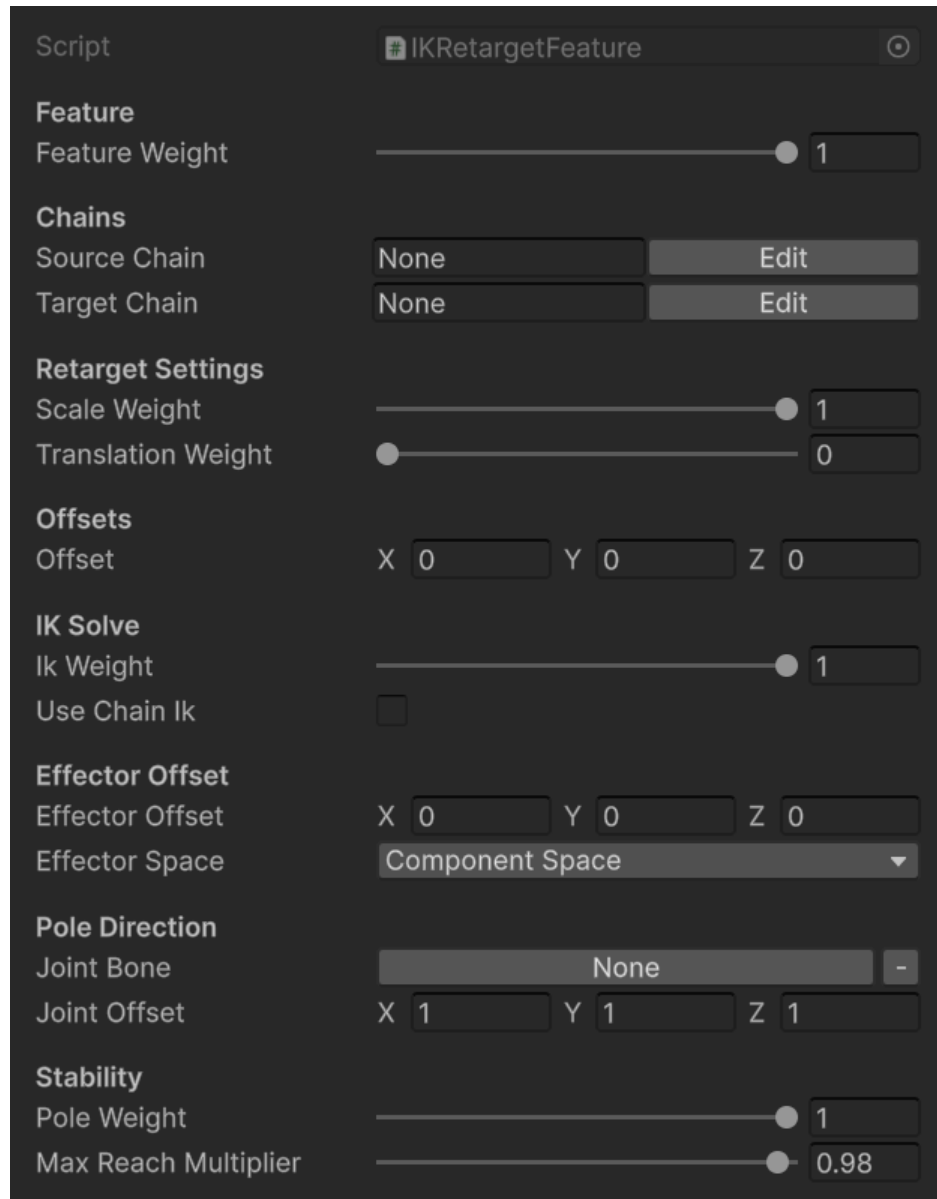


Copy Bone Inspector.

- **Copy From** — Source bones to copy motion from.
- **Copy To** — Target bones to copy motion to.

■ IK Feature

IK Feature is powered by inverse kinematics to accurately retarget motion. It's often used to retarget arms and legs chains.



The image shows a dark-themed UI panel titled "IK Retarget Inspector". At the top, there's a "Script" dropdown menu set to "IKRetargetFeature". Below this, the panel is organized into several sections:

- Feature**: A "Feature Weight" slider set to 1.
- Chains**: "Source Chain" and "Target Chain" are both set to "None", each with an "Edit" button.
- Retarget Settings**: "Scale Weight" slider is at 1, and "Translation Weight" slider is at 0.
- Offsets**: "Offset" section with input fields for X (0), Y (0), and Z (0).
- IK Solve**: "Ik Weight" slider is at 1, and "Use Chain Ik" is an unchecked checkbox.
- Effector Offset**: "Effector Offset" section with input fields for X (0), Y (0), and Z (0). "Effector Space" is a dropdown menu set to "Component Space".
- Pole Direction**: "Joint Bone" is set to "None" with a minus button. "Joint Offset" section with input fields for X (1), Y (1), and Z (1).
- Stability**: "Pole Weight" slider is at 1, and "Max Reach Multiplier" slider is at 0.98.

IK Retarget Inspector.

- **IK Weight** — Controls the influence of the IK.
- **Use Chain IK** — Toggles IK for long bone chains. It's recommended to keep it false for Humanoids, and true for Quadrupeds.
- **Effector Offset** — Position offset for the IK target.
- **Effector Space** — Which space the effector offset will be applied in.
- **Joint Bone** — hint or pole target bone (e.g., elbow or a knee).
- **Joint Offset** — hint or pole target offset.

- Pole Weight — Controls the influence of the pole/hint target.
- **Max Reach Multiplier** — Max allowed stretch length.

■ Root Pelvis Feature

Root Pelvis Feature is used exclusively to retarget root and pelvis bones motion.

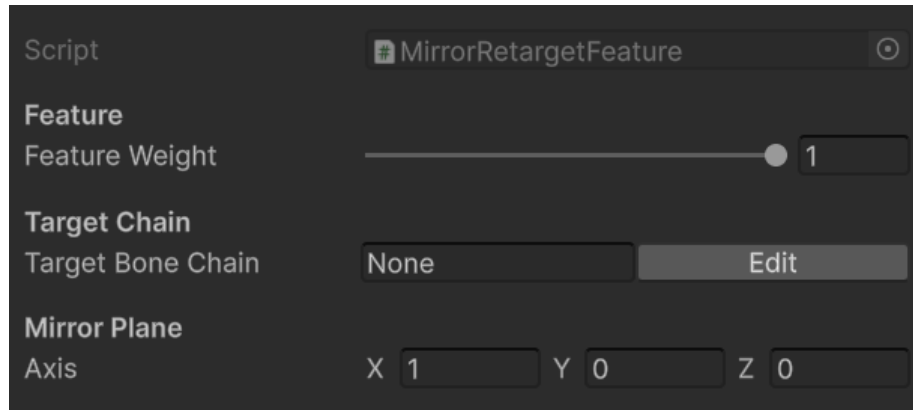


Root Pelvis Inspector.

- **Feature Weight** — Controls the influence of this feature.
- **Source Chain** — Source bones to retarget from.
- **Target Chain** — Target bones to retarget to.
- **Scale Weight** — Controls the influence of the body scale factor. If set to 1, it will be fully applied.
- **Translation Weight** — Controls the influence of the translation factor.
- **Offset** — Translation offset property.

■ Mirror Feature

Mirror Feature is used to mirror a character pose along the specified axis.



Mirror Feature Inspector.

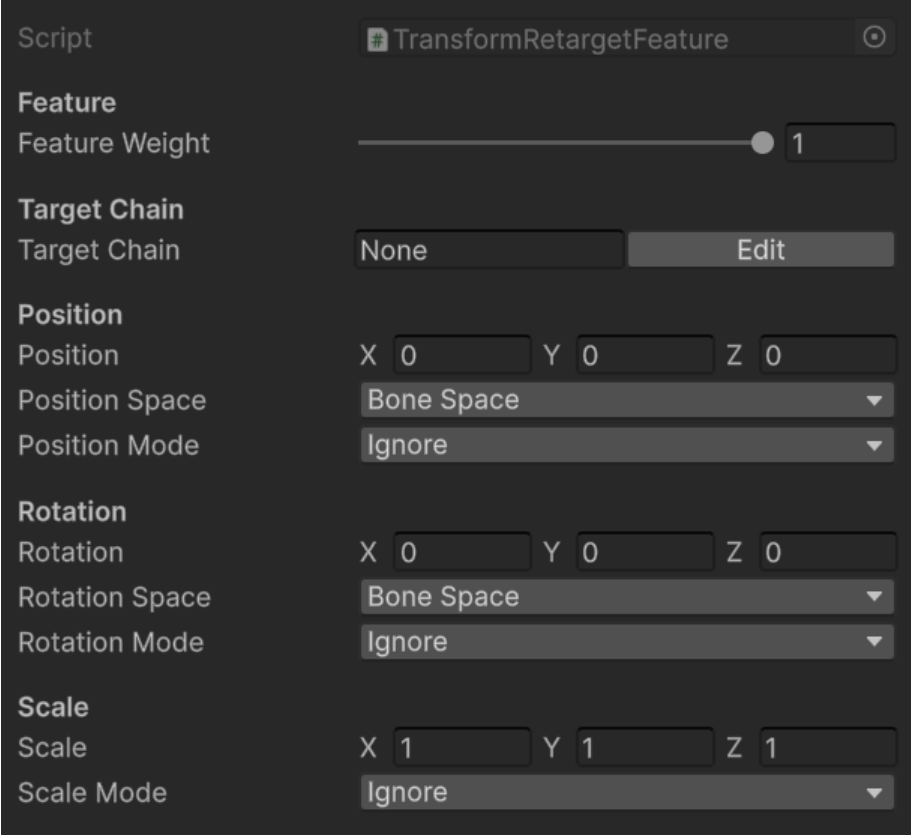
- **Target Bone Chain** — Bones that will be mirrored.
- **Axis** — Mirroring axis for target bones.

INFO

Note: By default, the character pose will be mirrored vertically.

■ Transform Feature

Transform Feature modifies selected bone chain position, rotation and scale.

The image shows a dark-themed user interface for the 'Transform Feature Inspector'. At the top, a 'Script' dropdown is set to 'TransformRetargetFeature'. Below this, the 'Feature' section contains a 'Feature Weight' slider set to 1. The 'Target Chain' section has a 'Target Chain' dropdown set to 'None' and an 'Edit' button. The 'Position' section includes 'Position' (X: 0, Y: 0, Z: 0), 'Position Space' (Bone Space), and 'Position Mode' (Ignore). The 'Rotation' section includes 'Rotation' (X: 0, Y: 0, Z: 0), 'Rotation Space' (Bone Space), and 'Rotation Mode' (Ignore). The 'Scale' section includes 'Scale' (X: 1, Y: 1, Z: 1) and 'Scale Mode' (Ignore).

Script	TransformRetargetFeature		
Feature			
Feature Weight	<input type="range" value="1"/>		
Target Chain			
Target Chain	None	Edit	
Position			
Position	X 0	Y 0	Z 0
Position Space	Bone Space		
Position Mode	Ignore		
Rotation			
Rotation	X 0	Y 0	Z 0
Rotation Space	Bone Space		
Rotation Mode	Ignore		
Scale			
Scale	X 1	Y 1	Z 1
Scale Mode	Ignore		

Transform Feature Inspector.

- **Position** — Vector translation property.
- **Position/Rotation Space** — Space in which the offset will be applied.
- **Position/Rotation/Scale Mode** — Add, Replace or Ignore modifications.

This feature is usually added to modify a specific bone or a chain after the main retargeting features. For example, it can be used to rotate the spine or a tail.

■ FPS Retargeting

FPS Feature retargets first-person animations. Unlike the standard [IK Feature](#), FPS Retarget positions the hands relative to the weapon, which results in a more accurate and consistent pose.

The screenshot shows the configuration panel for the 'FPSRetargetFeature' script. It is organized into several sections: 'Feature' with a 'Feature Weight' slider set to 1; 'Source Chains' with buttons to edit 'Source Right Arm', 'Source Left Arm', and 'Source Weapon'; 'Target Chains' with buttons to edit 'Target Right Arm', 'Target Left Arm', and 'Target Weapon'; 'IK Offsets' with input fields for 'Right Hand Offset', 'Left Hand Offset', and 'Weapon Offset' (each with X, Y, and Z components, all set to 0); and 'Pole IK Offsets' with input fields for 'Right Pole Offset' and 'Left Pole Offset' (each with X, Y, and Z components, all set to 0).

Script	FPSRetargetFeature
Feature	
Feature Weight	1
Source Chains	
Source Right Arm	Edit Source Right Arm
Source Left Arm	Edit Source Left Arm
Source Weapon	Edit Source Weapon
Target Chains	
Target Right Arm	Edit Target Right Arm
Target Left Arm	Edit Target Left Arm
Target Weapon	Edit Target Weapon
IK Offsets	
Right Hand Offset	X 0 Y 0 Z 0
Left Hand Offset	X 0 Y 0 Z 0
Weapon Offset	X 0 Y 0 Z 0
Pole IK Offsets	
Right Pole Offset	X 0 Y 0 Z 0
Left Pole Offset	X 0 Y 0 Z 0

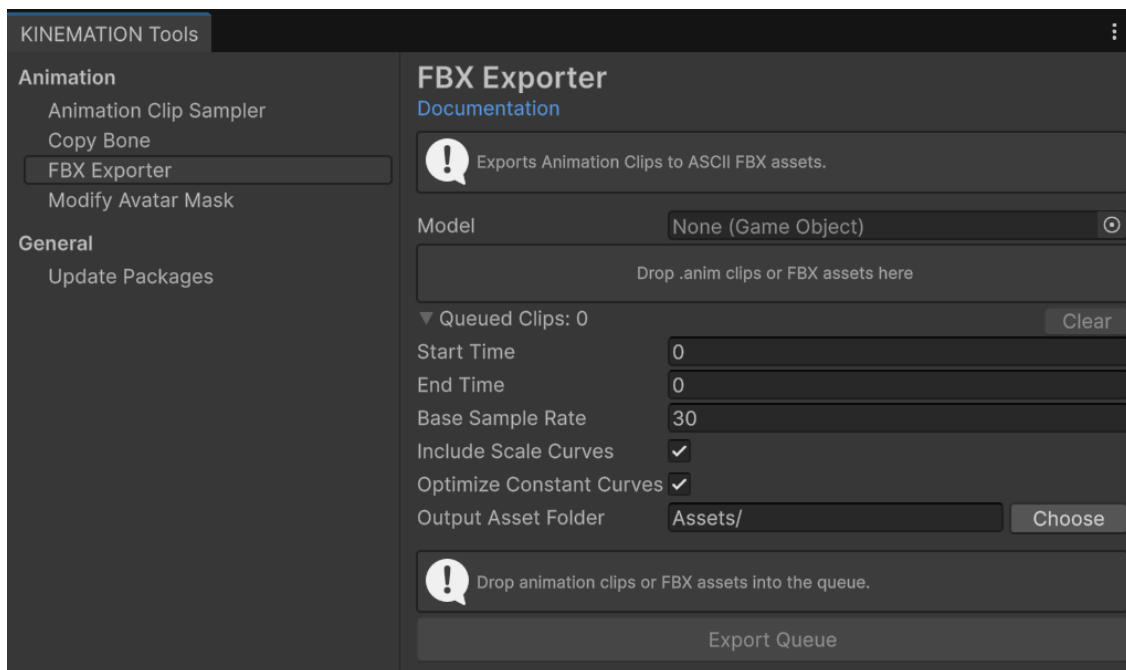
- **Source Right/Left Arm** — Right and left arm bones to retarget motion from.
- **Source Weapon** — Gun bone to retarget from.
- **Target Right/Left Arm** — Right and left arm bones to retarget motion to.
- **Target Weapon** — Gun bone to retarget to.
- **Right/Left Hand Offset** — Right/Left hand translation offset relative to the weapon.
- **Weapon Offset** — Weapon translation offset.
- **Right/Left Pole Offset** — Right and left elbow position offsets.

FBX Exporter

FBX Exporter is a built-in KINEMATION tool that lets you export animations as ASCII FBX files. It works with any skeleton and properly supports root motion.

How to use it

To use the tool, go to **Tools** ▶ **KINEMATION** ▶ **FBX Exporter**:



FBX Exporter settings.

- **Model** — the skinned mesh that will be animated.
- **Animation Clip** — the animation to export.
- **Start Time** — start time in seconds.
- **End Time** — end time in seconds.
- **Base Sample Rate** — target export frame rate (frames per second).
- **Include Scale Curves** — whether to include scale curves or not.
- **Optimized Constant Curves** — whether to remove redundant keyframes for constant curves.
- **Output Asset Folder** — where the exported FBX files will be saved.

Select a model and animation to export as FBX. Next, hit the "**Export FBX Animation**" button. Once exported, FBX info will be displayed under the button:

Export FBX Animation

Last Export

Assets/A_Locomotion_Stand_Jog_F_Loop_IP.fbx

Imported Clips2

Ping Exported Asset